Novel Neuroevolution to Train Neuromorphic AI using Co-Adversidal RL, P-bits, and Optimizations

Background

- Conventional AI training methods have stagnated and face efficiency issues (Figure 1 and Figure 2).
- New research on spiking neural networks (SNNs) shows they use binary neuron spiking, offering better efficiency.
- Traditional optimization relies on gradient-based methods, but SNNs are non-continuous, rendering this approach ineffective. Instead, other methods are used.
    - Surrogate gradient descent
    - Conversion of conventional neural weights to SNN format
    - STDP (spiking timing dependent plasticity) are used, where neurons self-optimize based on neighbors' behavior
- We propose a new training method to leverage SNNs' efficiency and to improve neural network performance. By implementing more biorealistic features with CPPNs (a function that generates a neural network). These CPPNs can be thought of as genomes that undergo evolution.

Background Visualization

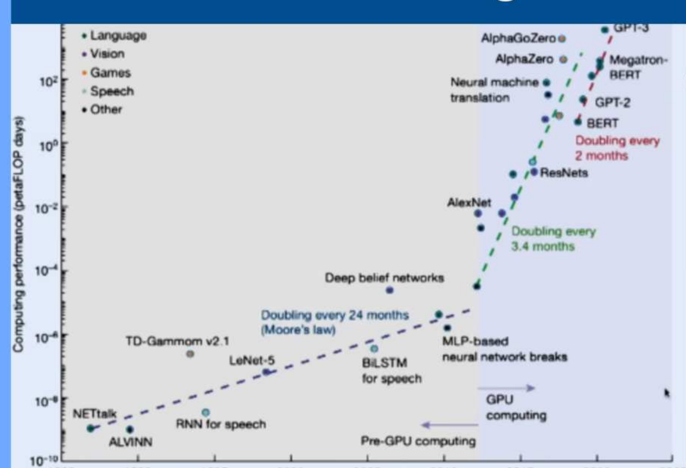


Figure 1: Compute required to train models over time. These trends were predictable at creating sizable increases in performance, but have since become outdated. URL:https://www.youtube.com/watch?v=VTKcsNrqdqA

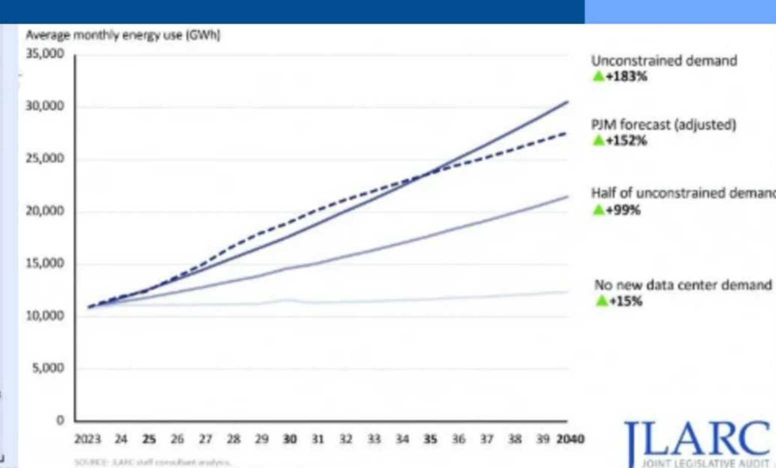


Figure 2: Amount of energy consumed in Virginia alone with respect to AI demand URL: http://jlarc.virginia.gov/landing-2024-data-centers-in-virginia.asp

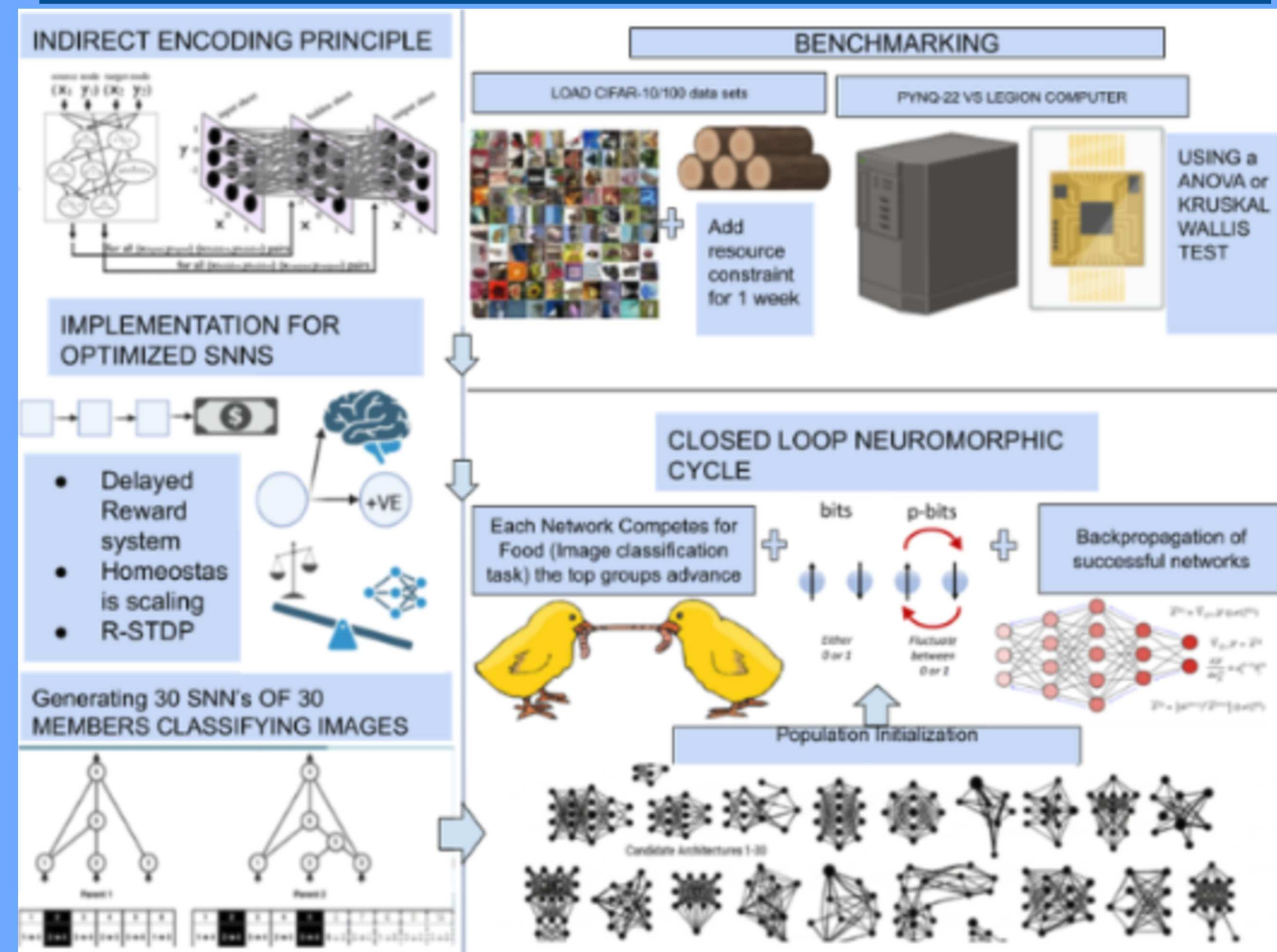
Hypothesis

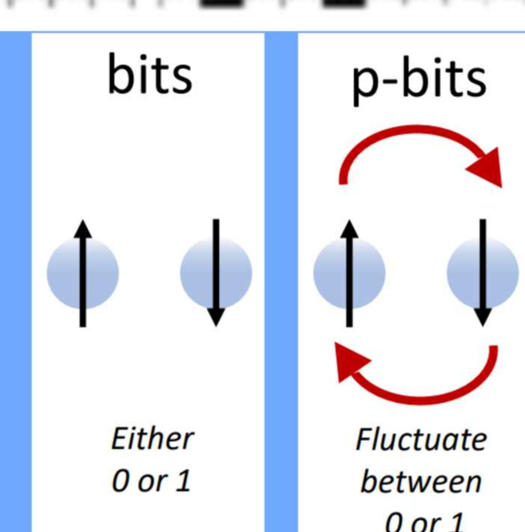
If we mimic biologically inspired brain features without its physical limits, we can design intricate, well-trained spiking neural networks that outperform conventional models. We challenge the assumption that gradient descent always finds the best solution given sufficient time, since "enough" is undefined in practice and its non-interpretablity. By combining this faster, more efficient architecture with training methods that leverage natural optimizations such as generalization and pruning, we can encourage better hardware development in this area and create smaller, more efficient AI systems that do not rely on intensive scaling laws while at least retaining if not increasing performance.

Purpose

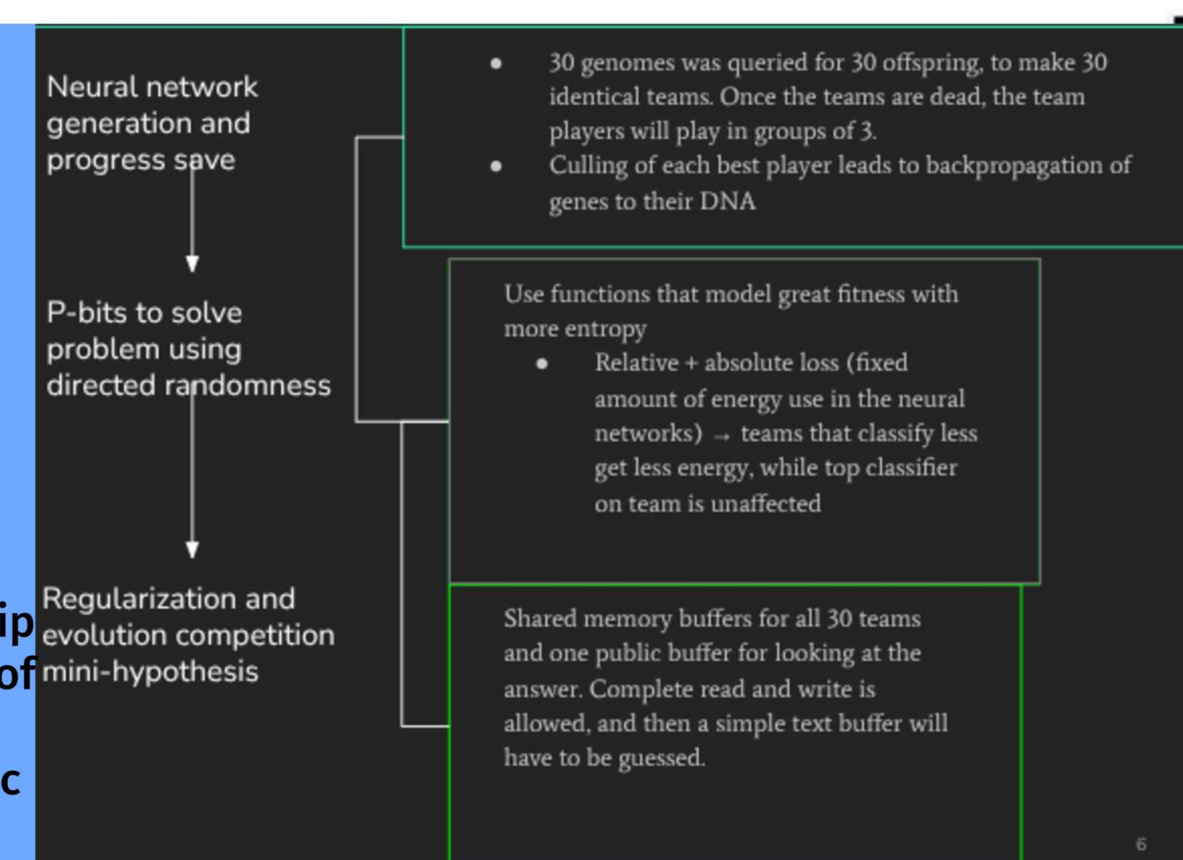
Newer AI models are producing less performance yields while requiring more resources. WE aim to reduce that

Methodology Visualization





Captions: P-bits are bits that randomly flip between 0 and 1 based on entropy of room-temperature (true randomness). URL:https://engineering.purdue.edu/semiconductors/research-areas



Methodology Reasoning Summary: Conventional methods of training AI is like a hiker going down a hill where the hill represents error and lower is better. This one hiker is expensive, but sophisticated and has bincoulars to track the steepest path to reduce the error of the model per training cycle. Our method is like if you had 30 teams of 30 lousy hikers who could commuicate with each other to converge on the lowest known spot. However, during their convergence, they would make deep discoveries that allow them to greatly optimize. Their stochasticity allows them to "research" the terrain and find the true lowest point given enough time, and they can logically assess based on their known knowledge the best combination of "coordinates" that are known for yielding good discoveries. Over time, they will make more and more discoveries as they move towards each other, and logically optimize for the lowest error. This computation will happen faster and cheaper as they operate off "free energy" in thermal fluncations and have multiple approximations / optimizations to make them lazily efficient.

Limitations

- The project was too ambitious:
    - It required extensive rewriting of complex low-level code for CUDA kernels responsible for LLM inference into a different format with a different architecture for HyperNEAT optimized inference.
    - ·It also required extensive writing of VHDL, a hardware descriptor language, to prototype our own hardware circuits using a PYNQ-Z2 to act as a simuation controller that could use the GPU in our provided Legion laptop as a surrogate for high performance compute while it mangaged asychrونous event handling and specialized compute paths that would make a GPU stutter.
    - We also would have to measure joules used by the PYNQ-Z2, on top of benchmarks of the evolved AI model loaded on it against conventional methods which is harder to do.
    - P-bit require extensive probablity equations based on thermodynamics and entropy, and we couldn't find enough to model that of a wild animal
    - There were multiple blockages to our research that required on-going research such as the viablity of neuroevolution versus a dedicated weight generator for connections between neurons, insights on evolutionary theory that is different compared to conventional methods, and modeling everything in a low-energy, optimized format through approximations or architectures.

Data + Results

None

Future Research

- Need to strip down optimizations and look at the core. We know p-bit equations out there exist, however they are hard to track, and just need to be found. Once this is done everything else falls into place, but we can't make these equations by ourselves at this time. We felt into a rabbit-hole that requires much more time than our project is given.
- Creating the simulation controller and constriants that best guided the candidates without giving them a chance for simulaton and unseen solutions that could break the simuiation also need to be patched.

References

Balduzzi, D., & Besserve, M. (2012). Towards a learning-theoretic analysis of spike-timing dependent plasticity (No. arXiv:1209.5549). arXiv. https://doi.org/10.48550/arXiv.1209.5549
Cruz-Duarte, J. M., & Talbi, E.-G. (2025). NeurO optimisation: The spiking way to evolve (No. arXiv:2507.08320; Version 1). arXiv. https://doi.org/10.48550/arXiv.2507.08320
Custode, L. L., Mo, H., Ferigo, A., & Iacca, G. (2022). Evolutionary optimization of spiking neural p systems for remaining useful life prediction. Algorithms, 15(3), 98. https://doi.org/10.3390/a15030098
Deng, S., Lin, H., Li, Y., & Gu, S. (2023). Surrogate module learning: Reduce the gradient error accumulation in training spiking neural networks. Proceedings of the 40th International Conference on Machine Learning, 7645–7657. https://proceedings.mlr.press/v202/deng23d.html
Florian, R. V. (2007). Reinforcement learning through modulation of spike-timing-dependent synaptic plasticity. Neural Computation, 19(6), 1468–1502. https://doi.org/10.1162/neco.2007.19.6.1468
Guo, W., Fouda, M. E., Eltawil, A. M., & Salama, K. N. (2021). Neural coding in spiking neural networks: A comparative study for robust neuromorphic systems. Frontiers in Neuroscience, 15. alduzzi, D., & Besserve, M. (2012). Towards a learning-theoretic analysis of spike-timing dependent plasticity (No. arXiv:1209.5549). arXiv. https://doi.org/10.48550/arXiv.1209.5549
Cruz-Duarte, J. M., & Talbi, E.-G. (2025). NeurO optimisation: The spiking way to evolve (No. arXiv:2507.08320; Version 1). arXiv. https://doi.org/10.48550/arXiv.2507.08320
Custode, L. L., Mo, H., Ferigo, A., & Iacca, G. (2022). Evolutionary optimization of spiking neural p systems for remaining useful life prediction. Algorithms, 15(3), 98. https://doi.org/10.3390/a15030098
Deng, S., Lin, H., Li, Y., & Gu, S. (2023). Surrogate module learning: Reduce the gradient error accumulation in training spiking neural networks. Proceedings of the 40th International Conference on Machine Learning, 7645–7657. https://proceedings.mlr.press/v202/deng23d.html
Florian, R. V. (2007). Reinforcement learning through modulation of spike-timing-dependent synaptic plasticity. Neural Computation, 19(6), 1468–1502. https://doi.org/10.1162/neco.2007.19.6.1468
Guo, W., Fouda, M. E., Eltawil, A. M., & Salama, K. N. (2021). Neural coding in spiking neural networks: A comparative study for robust neuromorphic systems. Frontiers in Neuroscience, 15. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin
Huizinga, J., Clune, J., & Mouret, J.-B. (2014). Evolving neural networks that are both modular and regular: HyperNEAT plus the connection cost technique. Proceedings of the 2014 Annual Conference on Genetic and Evolutionary Computation, 697–704. https://doi.org/10.1145/2576768.2598232
Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling laws for neural language models (No. arXiv:2001.08361). arXiv. https://doi.org/10.48550/arXiv.2001.08361
Liu, C., & Wang, H. (2025). Auto deep spiking neural network design based on an evolutionary membrane algorithm. Biomimetics, 10(8), 514. https://doi.org/10.3390/biomimetics10080514
Mead, C. (1990). Neuromorphic electronic systems. Proceedings of the IEEE, 78(10), 1629–1636. https://doi.org/10.1109/5.58356
Neftci, E. O., Mostafa, H., & Zenke, F. (2019). Surrogate gradient learning in spiking neural networks (No. arXiv:1901.09948). arXiv. https://doi.org/10.48550/arXiv.1901.09948
Risi, S., & Stanley, K. O. (2011). Enhancing ES-HyperNEAT to evolve more complex regular neural networks. Proceedings of the 13th Annual Conference on Genetic and Evolutionary Computation, 1539–1546. https://doi.org/10.1145/2001576.2001783
Risi, S., & Stanley, K. O. (2012). An enhanced hypercube-based encoding for evolving the placement, density, and connectivity of neurons. Artificial Life, 18(4), 331–363. https://doi.org/10.1162/ARTL_a_00071
Shouval, H. Z., Wang, S. S.-H., & Wittenberg, G. M. (2010). Spike timing dependent plasticity: A consequence of more fundamental learning rules. Frontiers in Computational Neuroscience, 4. https://doi.org/10.3389/fncom.2010.00019
URL For first image- https://jlarc.virginia.gov/landing-2024-data-centers-in-virginia.asp
URL for second image - https://www.youtube.com/watch?v=VTKcsNrqdqA

Huizinga, J., Clune, J., & Mouret, J.-B. (2014). Evolving neural networks that are both modular and regular: HyperNEAT plus the connection cost technique. Proceedings of the 2014 Annual Conference on Genetic and Evolutionary Computation, 697–704. https://doi.org/10.1145/2576768.2598232
Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling laws for neural language models (No. arXiv:2001.08361). arXiv. https://doi.org/10.48550/arXiv.2001.08361
Liu, C., & Wang, H. (2025). Auto deep spiking neural network design based on an evolutionary membrane algorithm. Biomimetics, 10(8), 514. https://doi.org/10.3390/biomimetics10080514
Mead, C. (1990). Neuromorphic electronic systems. Proceedings of the IEEE, 78(10), 1629–1636. https://doi.org/10.1109/5.58356
Neftci, E. O., Mostafa, H., & Zenke, F. (2019). Surrogate gradient learning in spiking neural networks (No. arXiv:1901.09948). arXiv. https://doi.org/10.48550/arXiv.1901.09948
Risi, S., & Stanley, K. O. (2011). Enhancing ES-HyperNEAT to evolve more complex regular neural networks. Proceedings of the 13th Annual Conference on Genetic and Evolutionary Computation, 1539–1546. https://doi.org/10.1145/2001576.2001783
Risi, S., & Stanley, K. O. (2012). An enhanced hypercube-based encoding for evolving the placement, density, and connectivity of neurons. Artificial Life, 18(4), 331–363. https://doi.org/10.1162/ARTL_a_00071
Shouval, H. Z., Wang, S. S.-H., & Wittenberg, G. M. (2010). Spike timing dependent plasticity: A consequence of more fundamental learning rules. Frontiers in Computational Neuroscience, 4. https://doi.org/10.3389/fncom.2010.00019
URL For first image- https://jlarc.virginia.gov/landing-2024-data-centers-in-virginia.asp
URL for second image - https://www.youtube.com/watch?v=VTKcsNrqdqA